B51TE Assignment #1

In whisky distillation, the working fluid undergoes a number of processes to reach the final condition which is then 'casked' and stored for a number of years (maturation) before bottling.

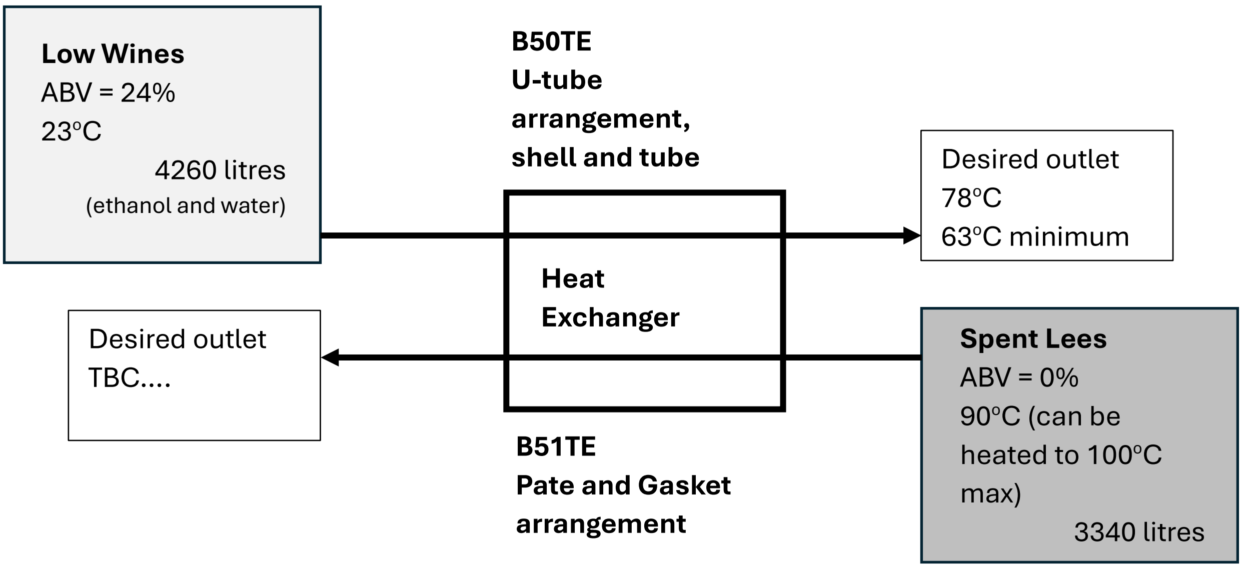
One such process occurs in the second distillation where products from the first distillation are distilled for a second time, after having been stored.

After the first distillation, 4260 litres of the alcohol (ethanol) bearing liquid (called Low Wines), which has an ABV (alcohol by volume) of 24% is stored at ambient conditions (23°C). 3340 litres of non-alcohol bearing product (called Spent Lees) is held in an insulated tank at 90°C.

Before the second distillation, the Low Wines are heated to 78°C. This can be done using heat recovered from the Spent Lees in a heat exchanger. The Low Wines must react at least 63°C from the heat recovery; 78°C is the target temperature.

You may make the following assumptions:

- The fluids will always be in the liquid phase.
- Pressure losses of 50 kPa are acceptable in both fluids; pumps are be used to move the fluids through the heat exchanger.
- An allowance for fouling should be accommodated.



Design a suitable plate and gasket heat exchanger that can be used for heat recovery and maximises the outlet temperature of the low wines. The Low Wines can be heated to the desired temperature using supplementary heating from the outlet heat exchanger temperature, if required.

Your final design should justify any supplementary heating requirements.

For the assignment,

Propose at least three possible designs and fully justify the final selection after comparing the merits of each design. [12 marks]

To achieve your final design you will have to experiment with different geometries and arrangements. Justify your final recommendation based on clearly defined, self-proposed criteria. You will also need to clearly state the decisions that you made and explain any observations that made from the software that led to final decisions.

For the chosen design, perform a manual heat balance (duty) to confirm the HTRI calculation. [2 marks]

Use your heat exchanger knowledge to replicate the duty calculation for the exchanger and remember to look at both flows.

For the Spent Lees, use existing correlations to manually estimate the heat transfer coefficient and compare with the HTRI value. Comment on any differences. [2 marks]

For the hot fluid, that is flowing in a channel, determine the heat transfer coefficient using a correlation of your choosing and comment on its value as compared to the value calculated by the software.

For the chosen design, use the heat transfer coefficients and tube information from HTRI to confirm the surface area by manual calculation (you may use the LMTD or E-NTU method) and comment on any difference between the manual and computer calculations.

[4 marks]

Use your heat exchanger knowledge to replicate the area calculation for the exchanger using the HTRI values for temperature, duty and heat transfer coefficients. Comment on the accuracy of the method used against the actual area proposed by the software.

Your submitted report (pdf) should be concise; use tables HTRI reports and drawings etc to summarise designs. Describe methodology used to determine 'best' design and comment on observations made whilst designing the exchanger.

Do not repeat calculations, show an example calculation and then summarise any repeat calculations in a table.

Include appropriate diagrams and charts that were used in the calculation/design. Reference any external resources. Do not use GenAI for any part of this assignment.

B51TE	**12 marks**	**2 marks**	**2 marks**	**4 marks**
Liquid – liquid exchanger	Design of plate and gasket HE using HTRI to specifications (Duty, pressure loss, fluids etc). Select best design on own criteria from 3 presented. Students will need to make decisions on a number of aspects of the HE design.	Calculate the HE duty and compare to HTRI. Remember that one flow is a mixture of two fluids.	The Spent Lees is essentially water; it flows in a channel. Calculate the heat transfer coefficient	Compare results chosen design against corrected LMTD or E-NTU methods for area and heat transfer coefficients.
	Make sure that any reports imported from HTRI are legible. Summarise in a table and include full reports in appendices. Same for tube layout and exchanger drawings	*Think about the calculations done in class and in the notes; do not take 10 pages to cover what can be done in 2!*		